

PID-Based Temperature Control System (ESP32 + DHT22 + Blynk)

A closed-loop PID temperature control system built with an ESP32 microcontroller. The system reads real-time temperature from a DHT22 sensor and adjusts a PWM-controlled heating element to maintain a user-defined setpoint, with live monitoring and remote setpoint control through a Blynk IoT dashboard. Developed and simulated in the Wokwi simulator inside VS Code with PlatformIO — no physical hardware required.

How It Works

- The DHT22 sensor measures the current temperature.
- The ESP32 calculates the error between the setpoint and the current temperature.
- A PID algorithm computes a correction value using Proportional, Integral, and Derivative terms.
- The correction value is sent as a PWM signal to the heating element (simulated with an LED).
- Current temperature, setpoint, and heater output are shown live on the Blynk dashboard, with history on a chart.

Components Used

Component	Purpose
ESP32	Main controller, runs the PID algorithm
DHT22	Measures current temperature (feedback sensor)
LED + Resistor	Simulates the heating element, driven by PWM
Blynk IoT	Live monitoring and remote setpoint control
Wokwi	Online simulator (replaces real hardware)

Circuit Connections

From	To
DHT22 VCC / SDA / GND	ESP32 3V3 / GPIO 15 / GND
ESP32 GPIO 25	Resistor → LED anode (heater control)
LED cathode	ESP32 GND

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Blynk Setup

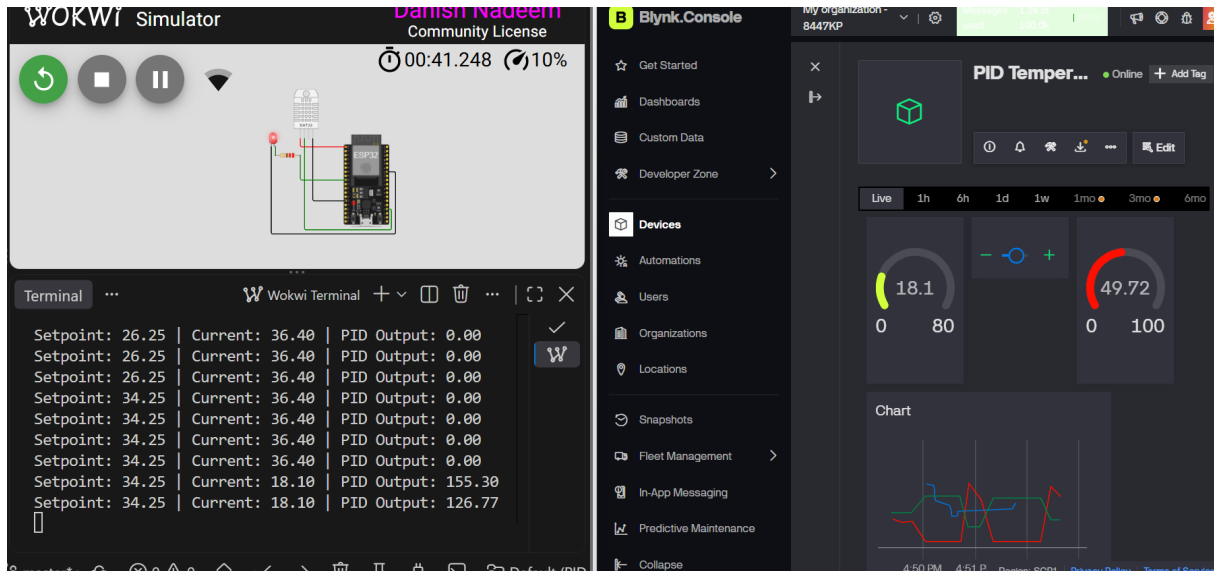
- Template: PID Temperature Controller (ESP32, WiFi).
- Datastreams: Current Temp (V0), Setpoint (V1), Heater Output (V2).
- Widgets: Gauge (V0), Slider (V1), Gauge (V2), Chart linking all three.

How to Demonstrate the Control Loop

Wokwi's simulated DHT22 does not automatically respond to the heater output, so the feedback loop is shown manually: set a target using the Setpoint slider, move the DHT22's temperature slider away from it, and watch the Heater Output rise. Move the DHT22 slider back toward the setpoint, and watch the Heater Output fall toward 0% as

the error closes.

Working Screenshot



Important Note

Never upload your real Blynk Auth Token to a public repository. Use placeholders in any shared code.

Tools and Technologies

ESP32, Arduino C/C++, PID Control Algorithm, Blynk IoT, Wokwi Simulator, PlatformIO, VS Code

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